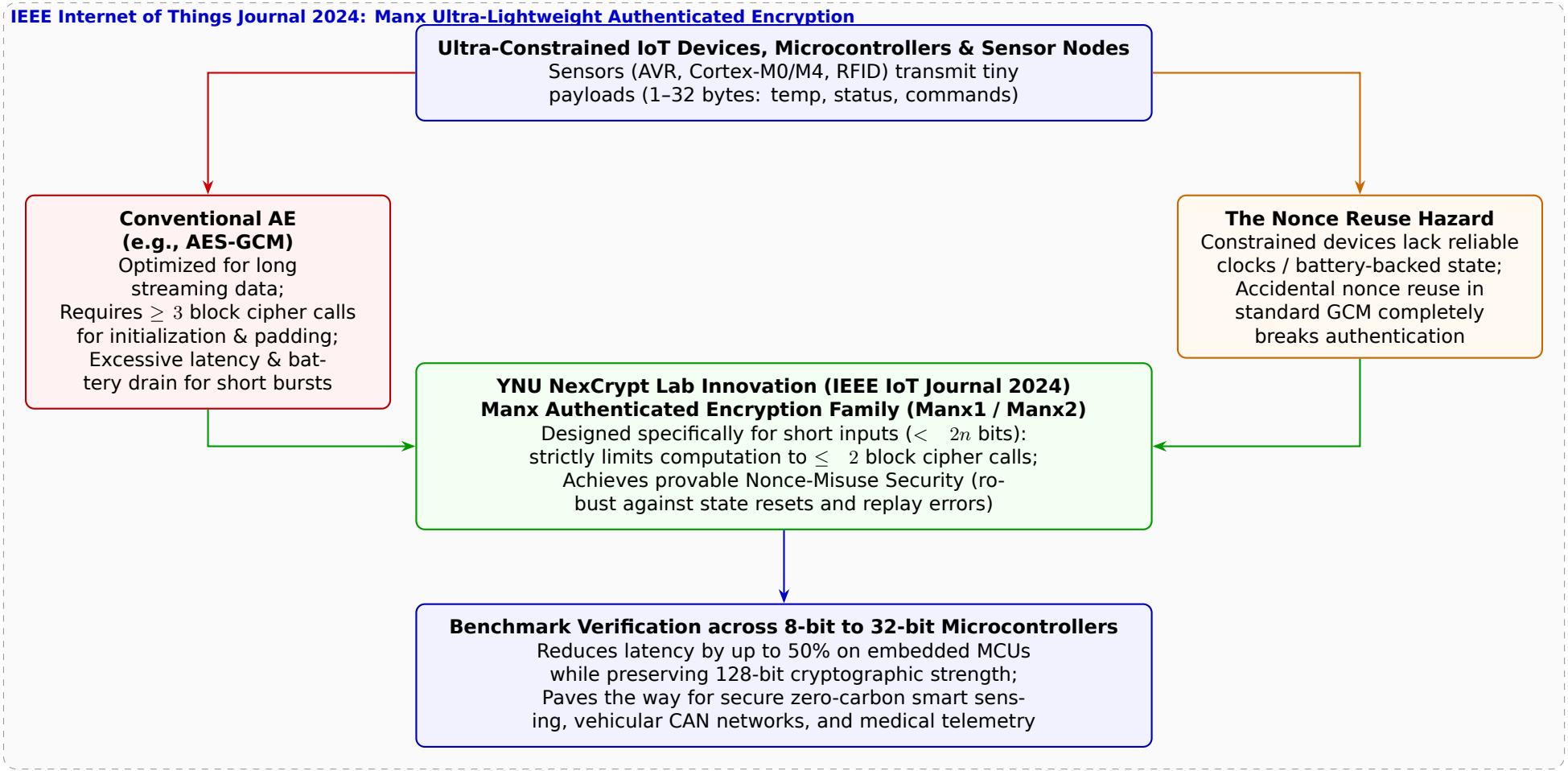

IEEE Internet of Things Journal 2024: Manx Ultra-Lightweight Authenticated Encryption
Ultra-Constrained IoT Devices, Microcontrollers & Sensor Nodes
Sensors (AVR, Cortex-M0/M4, RFID) transmit tiny payloads (1–32 bytes: temp, status, commands)
Conventional AE (e.g., AES-GCM)
Optimized for long streaming data; Requires ≥ 3 block cipher calls for initialization & padding; Excessive latency & battery drain for short bursts
The Nonce Reuse Hazard
Constrained devices lack reliable clocks / battery-backed state; Accidental nonce reuse in standard GCM completely breaks authentication
YNU NexCrypt Lab Innovation (IEEE IoT Journal 2024)
Manx Authenticated Encryption Family (Manx1 / Manx2)
Designed specifically for short inputs ($< 2n$ bits): strictly limits computation to ≤ 2 block cipher calls; Achieves provable Nonce-Misuse Security (robust against state resets and replay errors)
Benchmark Verification across 8-bit to 32-bit Microcontrollers
Reduces latency by up to 50% on embedded MCUs while preserving 128-bit cryptographic strength; Paves the way for secure zero-carbon smart sensing, vehicular CAN networks, and medical telemetry